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Theme: Smart Education

Task: Assignment 1

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Knowledge Representation

Our AI solution 'SimuLearn' is built with three core AI components, which include 'Intelligent Project Matching system', 'Guided Task Breakdown system' and 'Career Suggestion system'. This section will show the Knowledge Representation (KR) of each component.

1. Intelligent Project Matching

KR 1: Matching Syllabus Topic to Project Skill

- **Natural Language**

IF a student is learning a specific syllabus like 'Clustering', AND that topic covers a skill like 'Apply K-Means Algorithms', AND a project like 'Project_Alpha' requires the skill, THEN the project is recommended for the student.

- **First-Order Logic (FOL)**

$$\forall s, \forall p, \forall t, \forall k (IsLearning(s, t) \wedge CoversSkill(t, k) \wedge RequiresSkill(p, k) \rightarrow Recommends(p, s))$$

Explanation: This rule helps the system to achieve its goal "match skills to projects". This is the main logic that connects the syllabus that students learned, like 'Clustering', to a practical skill, like 'Apply K-Means Algorithms', then based on this skill to find a project that requires it. This rule is important to match what the student learned in their formal lectures to a real-world task. By creating this logical rule, the KR ensures the system can automatically recommend relevant content to the students. This fulfils the main user need that we need to solve, where students find it difficult to connect their learning to real application.

KR 2: Interpreting Free-Text Skill Input

- **Natural Language**

If a student enters a free-text description of their skills like "I can build a simple website that connected to database.", AND that text contains keywords (like "website", "database" and "simple"), THEN the system will assign a task like 'Web_Dev_Beginner' to that student.

- **First-Order Logic (FOL)**

$$\forall s, \forall t (HasInput(s, t) \wedge Contains(t, "website") \wedge Contains(t, "database") \wedge Contains(t, "simple") \rightarrow HasSkill(s, "Web_Dev_Beginner"))$$

Explanation: This rule also supports the "match skills" goal, but it makes the matching more personalized and flexible. It lets the AI understand informal, natural language descriptions, like "I built a simple website," and turn it into a structured skill that the system understands. This is important because not all student skills come from formal classes. This KR helps the system avoid suggesting projects that are too easy or too hard. It makes the system smarter by handling real-world, unstructured input, leading to a much more accurate and personalized project match for each student.

2. Guided Task Breakdown

KR 3: Enforcing Task Prerequisites

- **Natural Language**

If a student wants to start a specific task like 'Task_B', AND that task depends on a prerequisite like 'Task_A', BUT the student has NOT completed 'Task_A', THEN the student cannot start 'Task_B'.

- **First-Order Logic (FOL)**

$$\forall s, \forall tA, \forall tB (DependsOn(tB, tA) \wedge \neg IsComplete(s, tA) \rightarrow \neg CanStart(s, tB))$$

Explanation: This rule ensures that students complete all required tasks before proceeding to the next task to make sure that students follow a progressive learning flow where each task becomes more complex than the previous one. It is used to control access to higher-level tasks. If a student skips a prerequisite task, they may lack the foundational understanding needed for the next level. Therefore, completing prerequisite tasks is required to “level up” and unlock more challenging parts.

KR 4: Enabling Task Start

- **Natural Language**

If a student wants to start a specific task like “Task_B”, AND the task depends on other prerequisite tasks like “Task_A”, AND the student has completed all those prerequisite tasks, THEN the student is allowed to start “Task_B”. However, IF the task does not depend on any other tasks, students can start it immediately.

- **First-Order Logic (FOL)**

$$\forall s, \forall tA((\neg \exists tB(DependsOn(tA,tB)) \vee \forall tB(DependsOn(tA,tB) \rightarrow IsComplete(s,tB)))$$

Explanation: This rule is working together with Rule 3 which is Enforcing Task Prerequisites to manage overall project flow .It defines two conditions under which a task becomes “unlocked” for student.The first one is all prerequisite tasks have been successfully completed by the student.The second one is either the task is a starting tasks with no dependencies. By using these rules ,the system will be able to ensure that tasks are available in logical order,and prevent skipping work.

3. Career Suggestion System

KR 5: Identifying Task Preference Patterns

- **Natural Language**

If a student completes a task, AND the task is under a specific category such as 'Database', THEN the student's profile counter will be incremented for that category.

- **First-Order Logic (FOL)**

$$\forall s, \forall t, \forall c (IsComplete(s, t) \wedge HasCategory(t, c) \rightarrow IncrementProfile(s, c))$$

Explanation: This rule is a special type of rule which represents the AI's observation. It observes what task the student typically does and gathers data from the student's work. The data then will be used by the AI to make a suggestion tailored to the student preferences. It logs the type of tasks the student is successfully finishing. Hence, the system will be tailored personally according to the student work preferences and patterns.

KR 6: Making a Career Suggestion

- **Natural Language**

If a student's profile counter is high for a particular category like 'Database', AND their counter is low for another category like 'UI/UX Design', THEN the student will be suggested a suitable career that is related to the high count category like 'Database Administrator'.

- **First-Order Logic (FOL)**

$$\forall s (ProfileCount(s, "Database") > Threshold \wedge ProfileCount(s, "UI/UX Design") < Threshold \rightarrow SuggestCareer(s, "Database Administrator"))$$

Explanation: This rule works with the Rule 5 which gathered data of the student work and uses it to make a logical inference. It compares the student's preferences, and the result will be used to provide a personalized and relevant career suggestion to the student. Thus, these rules can fulfill one of the project's core goals.

Knowledge Representation Involved to Achieve Goals

The Knowledge Representation (KR) in *SimuLearn* serves as the fundamental that enables the AI system to connect between learning, practice, and career development of computing students. Each KR rule contributes to a different stage of the system's intelligent reasoning, allowing the system to behave more like a human tutor that can understand both a student's academic progress and real-world practicality.

The first two KRs (KR1 and KR2) focus on intelligent matching. It enables the AI to interpret what students describe about their learning and automatically link it with real-world projects that apply those skills. This allows students to directly learn and apply their lesson topics in actual company scenarios. Thus, the key problem of disconnected theoretical learning is addressed.

Besides, the next two KRs (KR3 and KR4) control the guided learning flow, preventing students from skipping the fundamental tasks that are given in a logical sequence. It guarantees gradual learning, prevents frustration, and supports mastery-based progression by building a solid foundation by enforcing preconditions and releasing the new challenges only when the student is ready.

Last but not least, KR5 and KR6 build a personalized career recommendations model. The AI will identify the student's strength and provide meaningful career guidance by analyzing the types of tasks a student prefers and performs well in. They transform the learning data into valuable insights to encourage and suggest students pursue careers aligned with their skills.

In summary, these Knowledge Representations allow *SimuLearn* to act as an intelligent connection between academic knowledge and career readiness. They enable the system the ability to reason, adjust, and customise learning experiences, which satisfies its ultimate objective of using intelligent, personalized education to prepare undergraduates for challenges in the real world of industry.